

Experiment 5

Question

Perform basic Image Handling and processing operations on the image. Read an image in Python and erode an image using the Erode function.

Aim

To read an image using Python and OpenCV and apply the erosion operation to shrink the bright regions of the image.

Procedure

1. Import the OpenCV (cv2) and NumPy libraries.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Create a kernel (structuring element) using NumPy.
6. Apply the erosion operation using cv2.erode().
7. Display the eroded image.
8. Save the eroded image.

Code

```
import cv2

import numpy as np

# Read image

image = cv2.imread("mickey.jpg")

# Check whether image is loaded

if image is None:

    print("Image not found!")

else:

    # Display original image

    cv2.imshow("Original Image", image)

# Create kernel

kernel = np.ones((5,5), np.uint8)

# Apply erosion

eroded = cv2.erode(image, kernel, iterations=1)

# Display eroded image

cv2.imshow("Eroded Image", eroded)

# Save output image

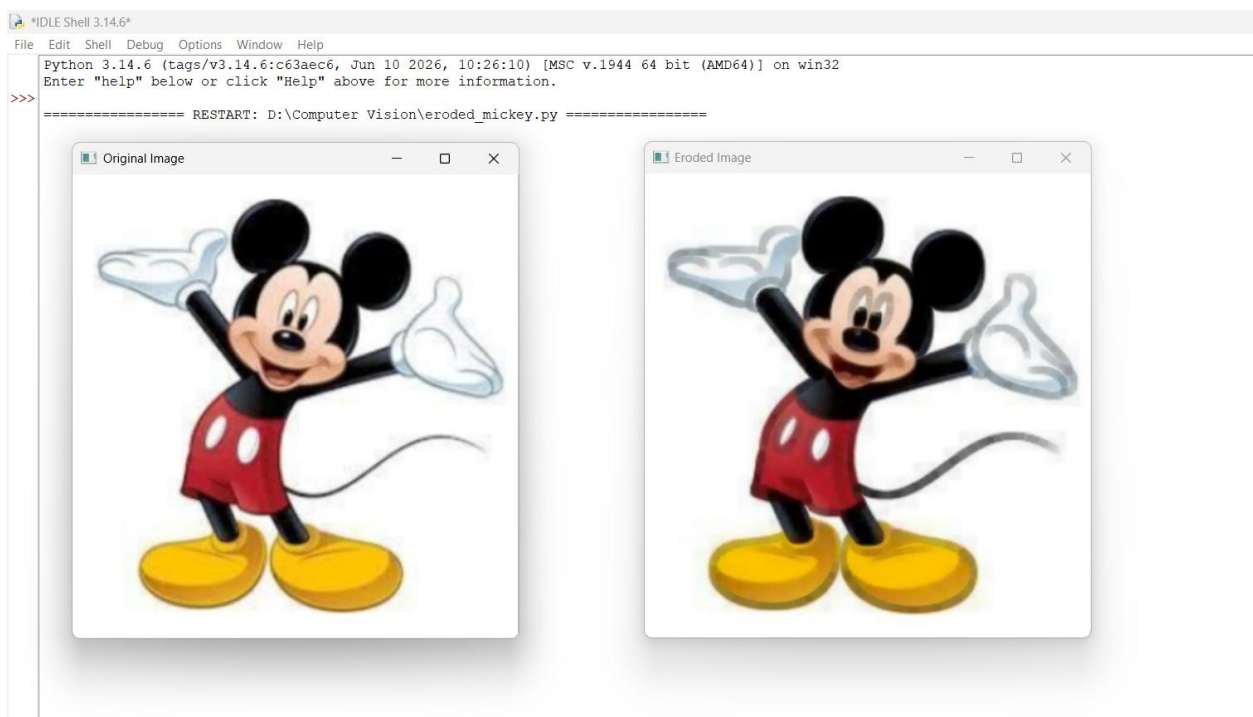
cv2.imwrite("eroded_mickey.jpg", eroded)
```

Input

Input Image: mickey.jpg

Output

1. **Original Image** : Displays original Mickey image.
2. **Eroded Image** : Displays image after applying the erosion operation.
3. **Saved File** : eroded_mickey.jpg is created in the project folder.



Result

The image was successfully read using OpenCV, the erosion operation was applied, and the eroded image was displayed and saved successfully.